

Part I: Choose the correct answer from the given alternatives (15 pts). put your Answer in the shaded region only.!!!!

- C** 1. A magnetic field exerts a force on a charged particle:
 A. always
 B. never
 C. if the particle is moving across the field lines
 D. if the particle is moving along the field lines
 E. if the particle is at rest
- C** 2. It is more difficult to measure the coefficient of volume expansion of a liquid than that of a solid because:
 A. no relation exists between linear and volume expansion coefficients
 B. a liquid tends to evaporate
 C. a liquid expands too much when heated
 D. a liquid expands too little when heated
 E. the containing vessel also expands
- A** 3. Heat is:
 A. energy transferred by virtue of a temperature difference
 B. energy transferred by macroscopic work
 C. energy content of an object
 D. a temperature difference
 E. a property objects have by virtue of their temperatures
- E** 4. The heat capacity of an object is:
 A. the change in its temperature caused by adding 1 J of heat
 B. the amount of heat energy that changes its state without changing its temperature
 C. the amount of heat energy per kilogram that raises its temperature by 1°C
 D. the ratio of its specific heat to that of water
 E. the amount of heat energy that raises its temperature by 1°C
- A** 5. During the time that latent heat is involved in a change of state:
 A. the temperature does not change
 B. the substance always expands
 C. a chemical reaction takes place
 D. molecular activity remains constant
 E. kinetic energy changes into potential energy
- D** 6. In an adiabatic process:
 A. the energy absorbed as heat equals the work done by the system on its environment
 B. the energy absorbed as heat equals the work done by the environment on the system
 C. the energy absorbed as heat equals the change in internal energy
 D. the work done by the environment on the system equals the change in internal energy
 E. the work done by the system on its environment is equal to the change in internal energy
- B** 7. The coefficient of linear expansion of iron is 1.0×10^{-5} per $^{\circ}\text{C}$. By what amount the surface area of an iron cube, with an edge length of 5.0 cm, will increase if it is heated from 10°C to 60°C ?
 A. 0.0125 cm^2 B. 0.025 cm^2 C. 0.075 cm^2 D. 0.15 cm^2 E. 0.30 cm^2
- B** 8. Faraday's law states that an induced emf is proportional to:
 A. the rate of change of the magnetic field
 B. the rate of change of the magnetic flux
 C. the rate of change of the electric field
 D. the rate of change of the electric flux
 E. none
- D** 9. What type of semiconductor results by doping silicon with trivalent atoms?
 A. N-type B. Intrinsic C. Superconductor D. P-type E. None

$$\beta = 2 \times 10^{-5}$$

$$A_0 = (5 \times 10^{-2})^2 = 25 \times 10^{-4}$$

$$\Delta A = \beta A_0 \Delta T$$

$$= 2 \times 10^{-5} \times 25 \times 10^{-4} \times 50^{\circ}\text{C}$$

$$= 2.5 \times 10^{-8} \text{ m}^2$$

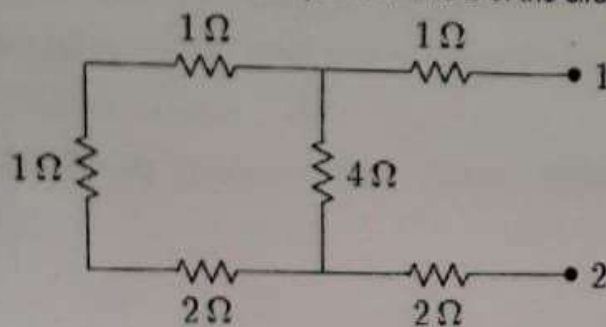
$$2.5 \times 10^{-8} \text{ cm}^2$$

$$0.0025$$

$$\frac{N \Delta \Phi}{\Delta t}$$

$$\frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

10. The equivalent resistance between points 1 and 2 of the circuit shown is:



- A. 3 Ω B. 4 Ω C. 5 Ω D. 6 Ω E. 7 Ω

Part II: Fill in the blank spaces with appropriate answers (10pts.) put your Answer in the shaded region only!!!!

- In **isobaric** process the pressure remains constant as other variables are changing.
- The temperature of a solid bar rises by 10°C when it absorbs 1.20 kJ of energy by heat. The mass of the bar is 500 g. The specific heat of the bar is **240 J/kg°C**.
- An object oscillates with simple harmonic motion along the x-axis. Its position varies with time according to the equation $x = (5.0 \text{ m})\cos\left(\frac{\pi}{2}t - \frac{\pi}{4}\right)$. Where t is in seconds and the angles in the parentheses are in radians. The period of the motion is **4 sec**.
- Transistor** is a semiconductor device used as a switch and an amplifier.
- A converging lens with 50cm focal length forms a real image that is 2.5 times larger than the object. The object distance from the image is **3.5m**.

$$Q = mc\Delta T$$

$$c = \frac{Q}{m\Delta T} = \frac{1.2 \times 10^3 \text{ J}}{0.5 \text{ kg} \times 10^{\circ}\text{C}} = 240$$

$$\omega = \frac{2\pi}{T} \quad \omega = \frac{\pi}{2}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

$$m = \frac{1}{2}$$

$$\frac{1}{2} = 2.5$$

Part IV. Solve the following problems by showing all the necessary steps (20 pts)

- For the circuit shown below, find
 - The currents in the three branches.
 - The potential difference between points b and d
 - The power dissipated across the $3\ \Omega$ resistor.

(5pts)

(a)

At Junction b,

$$I_1 + I_2 = I_3 \dots (1)$$

From loop a b d a,

$$9\text{ V} - 18 I_1 + 6 I_2 - 3\text{ V} = 0$$

$$6 - 18 I_1 + 6 I_2 = 0$$

$$18 I_1 - 6 I_2 = 6$$

$$3 I_1 - I_2 = 1 \dots (2)$$

From loop b c d b,

$$-3 I_3 + 3\text{ V} - 6 I_2 = 0$$

$$2 I_2 + I_3 = 1 \dots (3)$$

$$\begin{cases} I_1 + I_2 = I_3 \\ 3 I_1 - I_2 = 1 \\ 2 I_2 + I_3 = 1 \end{cases}$$

Substituting eq (1) + eq (3) gives

$$2 I_2 + (I_1 + I_2) = 1$$

$$I_1 + 3 I_2 = 1 \dots (4)$$

Solving eq (2) and eq (4) simultaneously,

$$\begin{array}{r} 3 I_1 - I_2 = 1 \\ + \quad I_1 + 3 I_2 = 1 \\ \hline 10 I_1 = 4 \end{array}$$

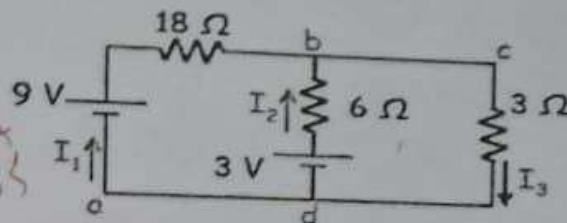
$$10 I_1 = 4$$

$$I_1 = 0.4\text{ A}$$

using this value in eq (2) gives

$$\begin{aligned} I_2 &= 3 I_1 - 1 \\ &= 3(0.4) - 1 \\ &= 0.2\text{ A} \end{aligned}$$

$$\text{using eq (1), } I_3 = I_1 + I_2 = 0.6\text{ A}$$



$$b) \Delta V_{bd} = V_d - V_b = ?$$

$$= 6 I_2 - 3\text{ V}$$

$$= 6(0.2) - 3\text{ V}$$

$$= 1.2\text{ V} - 3\text{ V}$$

$$= -1.8\text{ V}$$

$\therefore$ "d" is at lower potential than "b".

c)

$$P_{3\Omega} = I_3^2 R$$

$$= (0.6\text{ A})^2 (3\Omega)$$

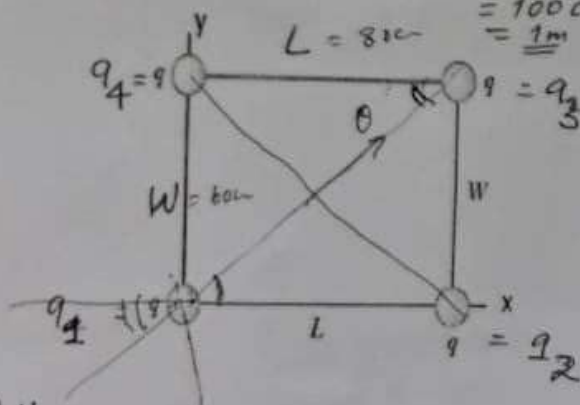
$$= 1.08\text{ W}$$

2. Four identical point charges $q = 10 \mu\text{C}$ are located on the corners of a rectangle as shown in the Figure below. The dimensions of the rectangle are $L = 80 \text{ cm}$ and $W = 60 \text{ cm}$. (Use: $k = 9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$). Calculate, (6 pts.)

- The magnitude of the resultant electric force on the charge located at the origin.
- The electric field at the center of the rectangle
- The potential energy required to assemble the system of charges on the four corners.

Given: $q_1 = q_2 = q_3 = q_4 = q = 10 \mu\text{C}$
 $L = 80 \text{ cm} = 0.8 \text{ m}$
 $W = 60 \text{ cm} = 0.6 \text{ m}$
 $K = 9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$

$$D = \sqrt{80^2 + 60^2} = 100 \text{ cm} = 1 \text{ m}$$



a) $\vec{F}_{\text{net}1} = \vec{F}_{21} + \vec{F}_{31} + \vec{F}_{41} = ?$

$$\vec{F}_{21} = \frac{k q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

$$= \frac{9 \times 10^9 \times 100 \times 10^{-12}}{(0.8)^2} \text{ N } (-\hat{i}) = -1.4 (\hat{i}) \text{ N}$$

$$\vec{F}_{31} = \frac{k q_1 q_3}{r_{13}^2} \hat{r}_{13}$$

$$= \frac{9 \times 10^9 \times 100 \times 10^{-12}}{(1)^2} \text{ N at } \tan^{-1}(3/4) = 37^\circ$$

$$= 0.9 \text{ N at } 37^\circ \text{ SW}$$

$$\vec{F}_{41} = \frac{k q_1 q_4}{r_{14}^2} \hat{r}_{14}$$

$$= \frac{9 \times 10^9 \times 100 \times 10^{-12}}{(0.6)^2} (-\hat{j}) = -2.5 \hat{j} \text{ N}$$

$$R_x = F_{21x} + F_{31x} + F_{41x}$$

$$= -1.4 + 0.9 \cos 37^\circ + 0$$

$$= -1.4 - 0.72$$

$$= -2.12$$

$$R_y = F_{21y} + F_{31y} + F_{41y}$$

$$= 0 - 0.9 \sin 37^\circ - 2.5$$

$$= -0.54 - 2.5$$

$$= -3.04$$

$$\therefore R = \sqrt{R_x^2 + R_y^2} = \sqrt{(2.12)^2 + (3.04)^2}$$

$$= 3.71$$

b) Zero! The fields cancel out at the center!

c) $U = \frac{k q_1 q_2}{r_{12}} + \frac{k q_1 q_3}{r_{13}} + \frac{k q_1 q_4}{r_{14}}$

$$= \frac{k q^2}{L} + \frac{k q^2}{D} + \frac{k q^2}{W}$$

$$= \frac{k q^2}{L} + \frac{k q^2}{D} + \frac{k q^2}{W}$$

$$= 7.833$$

3. A train is moving parallel to a highway with a constant speed of 20.0 m/s. A car is traveling in the same direction as the train with a speed of 40.0 m/s. The car horn sounds at a frequency of 510 Hz, and the train whistle sounds at a frequency of 320 Hz. (4 pts.)
- When the car is behind the train, what frequency does an occupant of the car observe for the train whistle?
 - After the car passes and is in front of the train, what frequency does a train passenger observe for the car horn?

$$v_t = 20 \text{ m/s}$$

$$v_c = 40 \text{ m/s}$$

$$f_c = 510 \text{ Hz}$$

$$f_t = 320 \text{ Hz}$$

From Doppler's frequency shift

$$f' = f \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

Where v - is speed of sound in air
 v_o - speed of observer
 v_s - speed of source.

$$\text{a) } \therefore f' = 320 \left(\frac{343 + 40}{343 + 20} \right) = \underline{338 \text{ Hz}}$$

$$\text{b) } f' = 510 \left(\frac{343 + 20}{343 + 40} \right) = \underline{483 \text{ Hz}}$$

4. A gas is compressed at a constant pressure of 0.8 atm from 9.0 L to 2.0 L. In the process, 400 J of energy leaves the gas by heat. (3 pts.)

(a) What is the work done on the gas?

(b) What is the change in its internal energy?

Given: $P_{\text{atm}} = 0.8 \text{ atm} = 0.8 \times 1.01 \times 10^5 \text{ Pa} = 0.808 \times 10^5 \text{ Pa}$ 0.5
 $V_i = 9 \text{ L} = 9 \times 10^{-3} \text{ m}^3$
 $V_f = 2 \text{ L} = 2 \times 10^{-3} \text{ m}^3$ 0.5
 $Q = -400 \text{ J}$

a) $W = -P \Delta V$ Isobaric Work 0.5
 $= -0.808 \times 10^5 (2 - 9) \times 10^{-3}$ 0.25
 $= 5.656 \times 10^2$
 $= \underline{\underline{565.6 \text{ J}}}$ 0.25

b) $\Delta U = Q + W$ 0.5
 $= -400 \text{ J} + 565.6 \text{ J}$ 0.25
 $= \underline{\underline{165.6 \text{ J}}}$ 0.25